ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** 20 N	60 - 40.
7	**(d)** 3822 Pa	Area = 3.14 x 0.01 = 0.0314 square m; 120/0.0314 = 3822 Pa.
8	**(d)** spaces between particles	Particles of one fill the gaps.
9	**(b)** 36 g, 1 : 8	4 + 32 = 36; 4:32 = 1:8.
10	**(c)** 39 per cent	64/164 x 100 = 39.
11	**(b)** moves towards team A	Net force towards A.
12	**(a)** A is likely to be stronger	Sea supplies energy.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R states exactly why friction is a contact force.
14	**(a)** Both A and R are true and R is the correct explanation of A.	The sharp edge has a smaller area, so the pressure is greater.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Sugar particles occupy the inter-particle spaces of water.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Mass remains 3 kg (1); Weight = 30 / 6 = 5 N (1).	1+1
17	Area = 0.005 square metre (1); P = 50/0.005 = 10000 Pa (1).	1+1
18	Gas particles move rapidly in all directions (1); They spread uniformly and fill the container (1).	1+1
19	Oxygen = 36 - 4 = 32 g (1); Ratio = 4 : 32 = 1 : 8 (1).	1+1
20	Percentage = 37.5 per cent (1); Water = 62.5 g (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Friction = 30 N (1); With 20 N net force = 10 N backwards (1); The box slows down and stops (1).	1+1+1
22	Area = 0.01 square m; P = 50/0.01 = 5000 Pa (1); Stacked: weight 100 N (1); P = 10000 Pa (1).	1+1+1
23	-10 to 0: ice warms (1); 0 to 100: melting at 0 then water warms (1); At 100: boiling then steam warms to 110 (1).	1+1+1
24	Probably iron, an element (1); Magnetic and reacts with acid giving hydrogen (1); Could also be a mixture containing iron (1).	1+1+1
	OR: NaCl: 4 g undissolved (1); KNO3: 8 g undissolved (1); NaCl is more soluble (1).	1+1+1
25	Friction depends on the force pressing surfaces (1); Larger area but less pressure per unit area (1); These balance out (1).	1+1+1
	OR: Pressure gradient is steepest (1); Air rushes toward the centre (1); Rotation increases speed (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) 60 - 40 = 20 N in the direction of the pull (1). (ii) Moving, since the net force is not zero (1). (iii) Opposite to the direction of motion (1). (iv) 60 - 10 = 50 N (1); rolling friction is much smaller than sliding friction, so less force is wasted (1).	1+1+1+2
27	Independent: temperature of water (1); Dependent: time taken for ink to spread (1); Constants: volume, ink drop, beaker (1); Observation: spreads faster in warm water (1); Conclusion: higher temperature gives faster diffusion (1).	1+1+1+1+1
	OR: Prepare a mixture of iron and sulphur (1); Test with a magnet before heating (1); Heat strongly in a test tube (1); Black, non-magnetic product forms after heating (1); A compound is formed; mixtures retain properties (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Weight equals air resistance.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 200 N	600/3.
7	**(c)** 4000 Pa	Inversely proportional.
8	**(b)** change of state	Latent heat.
9	**(c)** 1 g and 8 g	Ratio 1 : 8.
10	**(c)** 16 per cent	20+60 = 80 g in 500 g.
11	**(d)** magnetic force	Acts at a distance.
12	**(c)** wind and waves are strongest near the eye wall	Danger.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition of an element.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Very small particles cannot scatter light, so there is no Tyndall effect.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Mass is constant because the quantity of matter is the same; only weight changes.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Friction is lower (1); Less force opposes the sliding (1).	1+1
17	They show cloud patterns over a large area (1); Help in predicting path and intensity (1).	1+1
18	Evaporation is slow (1); Little cooling effect (1).	1+1
19	They have a lower melting point than the metals (1); Allow easy joining by melting (1).	1+1
20	Dissolved air is driven out on boiling (1); Little gas redissolves on cooling (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ice: least energy, close and fixed (1); Water: medium (1); Steam: highest, far apart (1).	1+1+1
22	In diamond all electrons are held in fixed bonds (1); In graphite some electrons are free to move (1); Different arrangements (1).	1+1+1
23	Dissolve impure crystals in hot water (1); Filter the hot solution (1); Cool slowly to get pure crystals (1).	1+1+1
24	Less distance to fall (1); Gravity acts for a shorter time (1); Speed has less time to increase (1).	1+1+1
	OR: Helium is lighter than air (1); Surrounding air provides upward push (1); Air-filled balloon is not lighter (1).	1+1+1
25	It rises slightly (1); Dissolved particles hinder escape of water particles (1); More heat needed (1).	1+1+1
	OR: A homogeneous mixture (1); Evaporate and condense water (1); Distillation (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Pressure = force / area (1). (ii) Area = 0.02 square metre, P = 600 / 0.02 = 30000 Pa (1). (iii) P = 600 / 0.01 = 60000 Pa (1). (iv) A camel's wide feet spread its weight over a large area, so the pressure on the sand is small; a heel has a small area, so the pressure is large (2).	1+1+1+2
27	Pushing in reduces volume, pressure rises (1); More collisions with walls (1); Pulling out lowers pressure (1); Outside air enters the pump (1); Pump compresses air into tyre (1).	1+1+1+1+1
	OR: Increase = 1600 times (1); Particles are far apart in the gas (1); Strong attraction in liquid (1); Little space between liquid particles (1); Gas fills available space (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A is true; R is false because mass does not depend on gravity.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** A is true but R is false.	A is true; R is false: heat is used to overcome forces between particles.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(c)** A is true but R is false.	A is true; R is false: powder has a larger surface area.
6	**(d)** 60 kg	Mass is unchanged.
7	**(a)** 60 N upward	Balanced forces.
8	**(a)** Particles have the same kinetic energy but different arrangement	Same temperature, different state.
9	**(d)** 14	2 x 7.
10	**(c)** 25 per cent	100/400.
11	**(b)** apply a lubricant	Reduces friction.
12	**(d)** reduce the force of the surge	Wave barrier.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Greater pressure at the bottom needs a stronger and thicker wall.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Faster particle motion increases diffusion.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but malleability and ductility do not explain conduction.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Spread the load (1); More grip and less pressure on the road (1).	1+1
17	Wind lowers pressure above the roof (1); Higher pressure inside pushes it up (1).	1+1
18	Only particles at the surface with enough energy escape (1); It occurs only at the liquid surface (1).	1+1
19	Their outer arrangement is stable (1); Used where reaction is unwanted, as in bulbs (1).	1+1
20	The solution is more concentrated (1); More sugar per unit mass of solution (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Rivers carry dissolved minerals into the sea (1); Evaporation removes only water (1); Salts accumulate over time (1).	1+1+1
22	Reduces the area facing the wind (1); Reduces air resistance (1); Helps ride faster (1).	1+1+1
23	Feet: 700/0.04 = 17500 Pa (1); Skis: 700/0.4 = 1750 Pa (1); Skis reduce pressure to one-tenth (1).	1+1+1
24	Hot water particles move faster (1); Higher average kinetic energy (1); Cold water particles move slower (1).	1+1+1
	OR: Sodium: soft reactive metal (1); Chlorine: poisonous greenish gas (1); NaCl: harmless white crystalline solid (1).	1+1+1
25	Sugar = 60 g (1); New total = 60/0.12 = 500 g (1); Water added = 300 g (1).	1+1+1
	OR: Rub a steel needle in one direction with a magnet (1); It acquires magnetism (1); Test by picking pins (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) Diffusion (1). (ii) The beaker at 65 degrees C (1). (iii) Particles of matter are always in motion and there are spaces between them (1). (iv) At a higher temperature particles have more kinetic energy and move faster, so they mix more quickly (2).	1+1+1+2
27	Forces on the man: weight down, reaction up (1); Both equal to 700 N at rest (1); Constant speed: net force zero (1); Reading remains 700 N (1); Only acceleration changes reading (1).	1+1+1+1+1
	OR: Cow: area 0.04 square metre (1); Cow pressure = 4000/0.04 = 100000 Pa (1); Man = 800/0.04 = 20000 Pa (1); Cow's pressure is five times (1); Cow may sink in wet soil more easily (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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